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# ACADEMIC & SCHEDULE MANAGEMENT SYSTEM (ASMS)

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# Project Link

https://github.com/DarShan3115/TEAM23_CS253_ASMS.git

# OVERVIEW

- Introduction
- Need of the software
- Target Users
- Major Requirements
- DEMO
- Key Features
- Plans
- Problems
- Discussion
- Conclusion

# INTRODUCTION

Overview

- Academic & Schedule Management System (ASMS)
- A centralized platform for managing academic activities
- Integrates tasks, courses, and performance tracking

# NEED OF THE SOFTWARE

**Problem**

Students often struggle with managing academic tasks, including assignments, deadlines, attendance, and course tracking. Existing tools are fragmented and do not provide a unified platform.

**Solution**

The Academic & Schedule Management System (ASAM) provides a platform to manage all academic activities efficeiently.

# Target Users

- Student
- Faculty
- Administration

# MINIMUM REQUIREMENTS

- Device (Smartphone / Computer)
- Stable internet connection
- Software
- Modern Web browser
- Standard operating system

(Basic setup required for system functionality)

System Dependencies

- Backend server
- Database
- Authentication (OTP services

# Functional Requirements

- Role-based authentication (admin-controlled registration)
- Account management and registration handling
- Course request and management by professors
- Assignment submission and grading
- Attendance tracking using log files
- Announcement system
- Dashboard with academic analytics

# Architecture

**Frontend :**

- React + Tailwind CSS

**Backend Services :**

- Auth Service (Node.js)
- Academic Service (Django)
- Productivity Service (Go)
- Analytics Service (FastAPI)

# FRONTEND ( React + Tailwind )

- Built using React (Vite) for fast rendering and modular UI
- Follows Atomic Design
- 11 core pages: Dashboard, Courses, Calendar, Tasks, Performance, Discussion Portal

Tailwind CSS for:

- Responsive design (mobile + desktop)
- Consistent dark-themed UI
- State Management: Zustand, handles global auth state
- Key Advantages are reusable components, fast performance, clean and consistent UI, easy scalability

# BACKEND SERVICES

- Polyglot Microservices Architecture: Each service uses best-suited technology
- Communicate via REST APIs

**Core Services:**

- Auth Service (Node.js/Express) for user login & registration
  - OTP-based authentication
  - JWT token generation
- Academic Service (Django)Courses, enrollments, assignments
  - Attendance & grading management
  - Central academic data hub

# BACKEND SERVICES

- Productivity Service (Go/Gin)Task management & reminders
  - Assignment submissions
  - Discussion forums
  - Handles high concurrency
- Analytics Service (FastAPI)GPA trends & performance analysis
  - Attendance insights
  - Identifies at-risk students
- Additional Points:
  - Shared PostgreSQL database (ACID compliance)
  - Dockerized system for consistency
  - Scalable, modular, and high-performance

# KEY FEATURES

- Role-based dashboards for students, faculty, and admins
- Course enrollment and tracking
- Assignment submission workflow
- GPA and attendance analytics
- Secure discussion portal

# DEMO

# FUTURE DEVELOPMENT PLANS

- AI driven task management
- Graph - based grade analysis with detailed performance insights
- A cross-platform Android and iOS app extending this website’s functionality.

# CONCLUSION

The Academic & Schedule Management System (ASMS) helps in organizing academic activities like classes, schedules, and tasks in a simple and efficient way. This project shows how technology can reduce manual work, save time, and improve coordination between students and faculty. By working as a team of 9 members, we also learned teamwork, problem-solving, and practical implementation of concepts. Overall, this system can make academic management more smooth, structured, and user-friendly.

# THANK YOU